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Model for grain boundary sliding and its relevance to optimal structural superplasticity

Part 1 – Theory

K. A. Padmanabhan and J. Schlipf

An assessment of the experimental findings leads to the conclusion that optimal structural superplasticity results from grain/interphase boundary sliding–diffusion coupled flow. An analysis of the boundary sliding process is presented first. By suggesting that both regions I and IIa (lower stress range of region II) of superplastic flow result from sliding–diffusion coupled flow, and treating mesoscopic (cooperative) boundary sliding as the rate controlling mechanism for optimal superplasticity, the stress, temperature, and grain size dependences of the strain rate of deformation are predicted. The above equation is then related to the stress exponent n (the inverse of the strain rate sensitivity index m). An analysis for determining the true activation energy for the rate controlling process is presented. Expressions for the distribution of internal stresses arising from sliding and the boundary viscosity are derived, and the presence of an initial unsteady region, predicted in an earlier analysis, is shown to be a natural consequence of this approach.

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© 1996 The Institute of Materials. Manuscript received 2 June 1994; in final form 6 July 1995. Professor Padmanabhan is in the Department of Metallurgical Engineering, Indian Institute of Technology, Madras, India and the late Professor Schlipf was in the Institut für Metallkunde und Metallphysik, Rheinisch-Westfälische Technische Hochschule, Aachen, Germany.

Introduction

'Micrograined' or 'structural' superplasticity has been identified in metallic materials, different kinds of ceramics, intermetallics, and composites. Hazzledine¹ has concluded that grain/interphase boundary sliding is unlikely to be rate controlling at any stage of structurally superplastic flow. Conversely, Frost and Ashby² have emphasised that at very low stresses experimental data are sparse and so the rates of the different mechanisms are uncertain. To the best of our knowledge, no physically valid reasons have been advanced to explain why boundary sliding can not be the controlling factor for the rate of flow.

One of the present authors previously suggested^{3,4} that boundary sliding controlled the rate of optimal (regions I and IIa) superplastic deformation. This was followed by a phenomenological treatment of the same process by Beere.⁵ In recent times, the idea that sliding could control the rate of superplastic flow has gained in importance.^{6–10} Therefore, in the view of the present authors, a simple picture of the boundary sliding process should be presented, if possible without involving assumptions that will preclude its use in understanding superplastic flow in intermetallics and/or ceramics. This paper attempts to achieve this.

Contributions of dislocation motion and diffusional creep to optimal superplastic flow

Many investigators have considered the role of dislocation motion in superplastic flow.^{10–19} Some regard it to be unimportant in regions I and IIa of superplastic flow,^{8–10,14,15,20} but in two studies dislocation activity has been reported under optimal conditions.^{21,22} Superplasticity in intermetallics, metal matrix composites, fine grained ceramics, ceramic composites, and nanograin ceramics has also been found to be dominated by grain sliding.^{23–29}

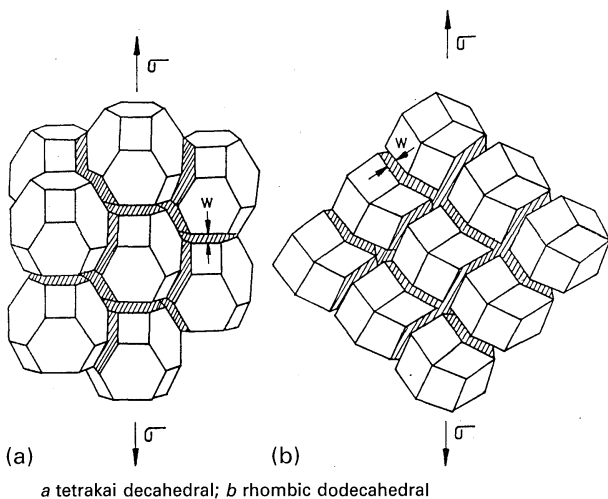
Following a detailed analysis of data, Valiev and Zelin³⁰ concluded that superplastic deformation in different classes of material appears to have a common mechanism of deformation and an argument may be made that the entire external strain is caused by grain boundary sliding.

However, evidence for limited amounts of non-rate controlling diffusion and dislocation motion in the optimal range of flow, is also present. Thus the case for rate controlling intragranular dislocation motion during optimal superplastic flow appears to rest on some early texture results.

The contemporary view^{31–36} is that the orientation distribution function data and analysis of Edington *et al.*^{12,15–17} (considered by the authors to be the most important piece of experimental evidence presented to date in favour of independent intragranular dislocation activity during optimal superplastic flow) were not sufficiently accurate or reliable and in addition in these experiments a non-optimal, extruded initial microstructure was used. A significant contribution from multiple slip was present only in regions IIb (the higher stress range of region II) and III of superplastic flow. The presence of significant grain rotations of an oscillatory nature both in sense and magnitude even in a single grain^{10,12} suggests the existence of unbalanced shear stresses at the grain and interphase boundaries, i.e. a case exists for regarding boundary sliding to be the rate controlling process.

Model for grain/interphase boundary sliding

Sliding and diffusive flow are interdependent processes and the slower of the two mechanisms will dictate the overall rate of deformation.^{37,38} In the Ashby–Verrall model³⁹ diffusion controls the rate of flow but in the present view^{4,10,40} grain/interphase boundary sliding is rate controlling. In this sense, these two approaches are extreme ways of looking at the same physical process. Both these approaches agree that regions I and IIa of superplastic deformation result from this physical process. Also, in agreement with most other theories, these two analyses consider the onset of relatively rate insensitive intragranular dislocation motion above a certain strain rate as the reason for the loss of superplasticity at high strain rates. This latter (non-optimal) region is not discussed further in this paper. Nix⁴¹ has noted that if diffusive flow were rate controlling as in the Ashby–Verrall model³⁹ no internal torque would be present and the experimentally observed grain rotations could not be accounted for.

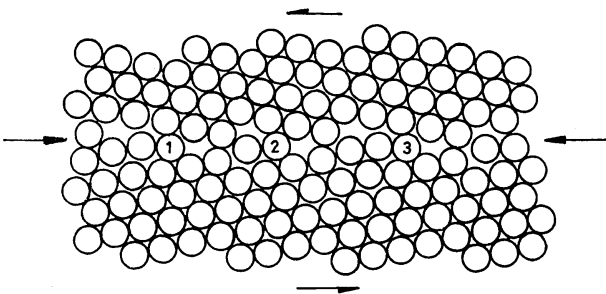


1 Schematic diagram⁴⁰ of rate controlling flow of grain boundaries surrounding arrays of essentially non-deforming (apart from what is needed for non-rate controlling accommodation between grains to ensure strain compatibility) grains during occurrence of optimal structural superplasticity: idealised shapes are only illustrative, as they are often suggested to be equilibrium and real shape of grains, respectively

Recently, Hazzledine and Schneibel⁴² have presented a general solution for the Coble diffusional creep problem in two dimensions in which grain rotations can take place by wedge plating. In that analysis, however, the rate of rotation of a given grain is constant, which is not consistent with the oscillatory grain rotations observed during superplastic flow.^{10,12} This problem may be overcome by making the rotation discontinuous and following the flow process iteratively.⁴³ As this has not yet been done, the relevance of this approach to the understanding of superplastic flow is not clear at this stage.

Conversely, Beere⁵ has already shown that when sliding controls the rate of flow, grain rotations of the right kind and switching of grain neighbours can be predicted. Recently, a three-dimensional sketch of the sliding controlled flow mechanism for idealised grain shapes has been presented (Fig. 1).⁴⁰ The rate controlling process is confined to the hatched network of deforming grain boundaries of width *W*. As already noted by Beere,⁵ some of the boundaries where sliding is most difficult may be bypassed by grain rotation and switching of neighbours. Within the actually deforming network of boundaries, flow in the region where sliding is most difficult will determine the overall rate of deformation. Outside the deforming boundaries, material transport by diffusion (which in the view of the present authors is not rate controlling) will be present in order to ensure coherency of deformation. A model for the rate controlling sliding process (a brief description of which was given at a recent conference⁴⁴) is proposed below.

It has not been possible to use the coincidence site lattice model for the structure of high angle grain boundaries⁴⁵ to predict quantitatively either the equilibrium or the non-equilibrium properties of grain boundaries. More recently, therefore, alternative models have emerged: a description analogous to that for liquids;^{46–50} the ‘structural unit model’;^{51–56} and that of Wolf⁵⁷ in which calculations show that high angle grain boundaries contain excess free volume of the order of 0.10 Ω , where Ω is the atomic volume in the lattice. This free volume is not distributed continuously along the boundary but is concentrated at certain sites depending on the orientation and misfit of the boundary. This point is illustrated by Fig. 2, after Pumphrey.⁵⁸ Grain boundary sliding occurs along such high angle boundaries during superplastic deformation. (It is emphasised that



2 Computed minimum energy structure for symmetric 24.4°/[111], $\Sigma 67$ boundary after Pumphrey;⁵⁸ boundary consists of many similar and not so similar atomic configurations, which are divided into many basic units of flow in present model

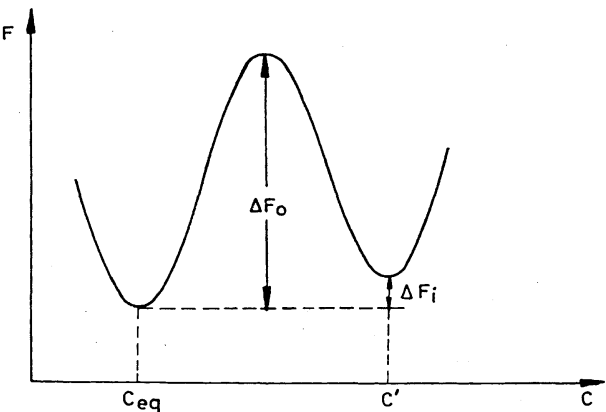
Fig. 2 is only illustrative. Moreover, as one approaches special boundaries and boundaries of low misorientation, the free volume may be a function of the boundary misorientation, thus giving rise to the experimentally observed orientation dependence of the sliding rate in this range. Treatment of this problem is beyond the scope of this paper.)

In the present model the atomic ensemble around a free volume site is considered as a basic unit of flow, e.g. the regions around sites 1, 2, and 3 in Fig. 2. The basic unit may be characterised by an area $A = \pi R_0^2$ in the plane of the boundary and a thickness *W* across the boundary (equal to the width of the grain boundary). More specifically, in order to be able to perform quantitative calculations, we take as a basic unit (on either side of the boundary) an oblate spheroid of volume

$$V_0 = \frac{4}{3} \pi R_0^2 \frac{W}{2} = \frac{2}{3} A_0 W \dots \dots \dots (1)$$

with $R_0 = W$. The main assumptions on which the model is based are as set out below.

1. The basic units of flow undergo shear transformations which are essentially independent of each other.
2. The ‘stress free’ shear within the basic unit, as brought about, for instance, by a thermally activated event, induces additional transient distortion inside and outside that basic unit. It is assumed that the elastic energy of these distortions constitutes the activation energy for the sliding process.
3. In the absence of external stress, the arrangement of atoms in the basic unit represents the stable equilibrium configuration C_{eq} (Fig. 3). It corresponds to an absolute minimum in the free energy $F(C_{eq})$. (If it is a non-equilibrium boundary, the initial arrangement of atoms in the basic unit would represent the lowest minimum among



3 Schematic diagram of relationship between free energy *F* and atomic configuration *C* for basic unit of boundary sliding

a series of local minima corresponding to many possible metastable configurations.) It is assumed that because of the presence of excess free volume in the basic unit there are atomic arrangements slightly different from C_{eq} which correspond to metastable configurations C' . These states have free energies

$$F(C') = F(C_{eq}) + \Delta F_i \text{ (see Fig. 3).}$$

Starting from C_{eq} they can be reached and converted into configurations C'_{eq} of free energy equal to C_{eq} (see below) by appropriate shear transformations γ_i provided that ΔF_i is not too large.

4. In the realistic case of a basic unit embedded in a solid matrix, a stress free shear would not be possible. It is assumed that concomitant with the internal shear a transient volume expansion occurs, which vanishes immediately on reaching the new state C'_{eq} . By contrast, the release of the induced shear component will depend on the progress of shear in the vicinity of the basic unit under consideration. It is only after the surroundings have been sheared by the same amount that the induced shear will be relaxed completely.

According to Eshelby⁵⁹ the elastic energy due to a shear γ_i in the basic unit is given by

$$\Delta F_1 = \frac{1}{2} \beta_1 \mu V_0 \gamma_i^2 \quad (2)$$

where μ is the shear modulus.

For an oblate spheroid (basic unit) with $W/2R_0 = 0.5$

$$\beta_1 = 0.470(1.590 - p)/(1 - p)$$

where p = Poisson's ratio. The elastic energy due to the concomitant volume expansion ε_i is given by

$$\Delta F_2 = \frac{1}{2} \beta_2 \mu V_0 \varepsilon_i^2 \quad (3)$$

where $\beta_2 = 4(1 + p)/[9(1 - p)]$.

The free energy of activation for an individual transformation is thus given by

$$\Delta F_0 = \Delta F_1 + \Delta F_2 = \frac{1}{2} (\beta_1 \gamma_i^2 + \beta_2 \varepsilon_i^2) \mu V_0 \quad (4)$$

(In Part 4 a way of estimating ΔF_0 is given.)

As shown schematically in Fig. 3, the difference $\Delta F_i = F(C') - F(C_{eq})$ produces a bias in favour of C_{eq} . It can be compensated for by an external shear stress $\tau \geq \tau_i$, where

$$\tau_i = \frac{\Delta F_i}{\gamma_i V_0} \quad (5)$$

represents an internal stress. For shear stresses $\tau \leq \tau_i$ no transition from C_{eq} to C'_{eq} will take place, since the rate of backward jumps will exceed the rate of forward jumps.⁴⁴ (That is, as the atoms approach the saddle point (see Fig. 3) the applied stress τ_i shears them into a configuration C'_{eq} , which has a free energy level equal to that of C_{eq} , i.e. the action of temperature and stress is simultaneous.)

According to Wolf⁵⁷ the excess free volume of high angle grain boundaries varies only slightly with misorientation. This has two consequences. First, for stresses $\tau \geq \tau_i$ the configuration C' is topologically equivalent to C_{eq} and forms the new equilibrium configuration C'_{eq} . Again there will be metastable states C'' nearby, which are transformed by the action of τ_i to a lower energy state of C'_{eq} equal to C_{eq} so that grain boundary sliding can proceed by a sequence of shear transformations. The equivalence of C_{eq} and C'_{eq} constitutes isoconfigurational flow kinetics which can be described by transition state theory.⁶⁰ Second, the individual shear transformations γ_i and volume expansions ε_i will not differ very much for different basic units or for the different metastable states within a basic unit. We assume, however, that the values of ΔF_i will vary because

the shear distortions will partly remain, depending on the state of deformation of the neighbourhood. Thus γ_i and ε_i will be replaced by their averages γ_0 and ε_0 , respectively, while a distribution $g(\tau_i)$ of the internal stresses τ_i will be taken into account. Thus the rate of grain boundary sliding $\dot{S}$ will be given by

$$\dot{S} = S_0 \omega_f(\tau, \tau_i) \quad (6)$$

where S_0 is the contribution to sliding of an individual basic unit, and ω_f is the effective number of jumps per second from C_{eq} to C'_{eq} . The relationship between S_0 and the basic shear γ_0 is

$$S_0 \approx W \gamma_0 \text{ (see Appendix 1)} \quad (7)$$

According to rate theory, the general transition frequency is given by

$$\omega = v \left[\exp \left(-\frac{\Delta F_0 - \Delta w}{kT} \right) - \exp \left(-\frac{\Delta F'_0 + \Delta w}{kT} \right) \right] \quad (8)$$

where ΔF_0 and $\Delta F'_0$ are the free energies of activation for forward and backward jumping, respectively, v is the thermal vibration frequency, and

$$\Delta w = \frac{1}{2} \tau \gamma_0 V_0 \quad (9)$$

represents the work done by the shear component of the applied stress along the activation path leading to the saddle point. This work is gained when the jump is in the forward direction and expended when the jump is in the backward direction. Evidently from Fig. 3

$$\Delta F'_0 = \Delta F_0 - \Delta F_i \quad (10)$$

At high temperatures and low stresses $\Delta F_i \ll kT$ and $\Delta w \ll kT$. For these conditions equation (8) may be rewritten as

$$\omega = \frac{v}{kT} (2\Delta w - \Delta F_i) \exp(-\Delta F_0/kT) \quad (11)$$

Employing the notation

$$\omega_0 = v \exp(-\Delta F_0/kT) \quad (12)$$

and substituting from equations (5) and (9) into equation (11) noting that $\gamma_i \approx \gamma_0$

$$\omega = \omega_0 \frac{\gamma_0 V_0}{kT} (\tau - \tau_0) \quad (13)$$

Here, if $\tau \leq \tau_i$, the rate of forward jumping would be negative, whereas in reality it should be zero. This can be remedied by multiplying equation (13) by the Heaviside step function

$$\theta(\tau - \tau_i) = \begin{cases} 0; & \tau \leq \tau_i \\ 1; & \tau > \tau_i \end{cases} \quad (14)$$

which gives

$$\omega_f = \omega \theta(\tau - \tau_i) \quad (15)$$

The distribution function $g(\tau_i)$ is defined operationally, taking into account the momentary distribution of internal stresses which in turn depends on the momentary distribution of shear within the boundary. Therefore, $g(\tau_i)$ can not be determined absolutely but depends on the deformation parameters. Assuming $g(\tau_i)$ to be normalised according to

$$\int_0^\infty g(\tau_i) d\tau_i = 1 \quad (16)$$

then from equation (13)

$$\omega_f = \omega_0 \frac{\gamma_0 V_0}{kT} \int_0^\infty (\tau - \tau_i) \theta(\tau - \tau_i) g(\tau_i) d\tau_i \quad (17)$$

Since for $\tau \leq \tau_i$ the integrand of equation (17) will be zero, this may be rewritten as

$$\omega_f = \left(\frac{\omega_0 \gamma_0 V_0}{kT} \right) f(\tau) \quad \dots \quad (18)$$

where

$$f(\tau) = \int_0^\tau (\tau - \tau_i) g(\tau_i) d\tau_i \quad \dots \quad (19)$$

Finally, the total deformation rate due to grain boundary sliding is given by $\dot{\gamma} = \dot{S}/L$ where L is the grain size (see Appendix 1) from which

$$\dot{\gamma} = \left(\omega_0 \frac{W}{L} \frac{\gamma_0^2 V_0}{kT} \right) f(\tau) \quad \dots \quad (20)$$

is obtained.

Cooperative/mesoscopic sliding

Equation (20) predicts the sliding rate at a single boundary. After sliding for a length of the order of a grain side/diameter a triple junction is in the way and this terminates the motion. Thus steric hindrance renders this kind of boundary sliding relatively ineffective.

In order to produce substantial sliding on a mesoscopic scale, two or more grains must cooperate to form a plane interface which by interconnection with other similar plane interfaces enables long range sliding until stopped by an insurmountable barrier, e.g. an extra large grain or a coarse precipitate. In forming the plane interface grain boundary energy must be expended which is supplied by the external force. (Evidently the driving force for plane interface formation is the reduction in the overall grain boundary energy through a reduction in the grain boundary area in the direction of sliding. As can be seen below, it is suggested that the reduction in grain boundary area arises from local boundary migration for which clear experimental evidence exists. Moreover, this is a much easier process than that envisaged in diffusion models in which diffusion distances of the order of the grain size are involved.)

Thus, in the present approach grain/interphase boundary sliding is a two scale phenomenon. On the microscopic scale the mechanism described in the previous section operates and causes sliding at boundaries. Here, the obstacles to sliding are localised and can be overcome by thermal activation. Short range internal stresses develop at this level. On a mesoscopic scale, which may be characterised by the grain diameter, the obstacles to sliding are other grains or particles residing at grain/interphase boundaries. The internal stresses associated with this level are long range and are described as threshold stresses.

It should be noted that, unlike in conventional treatments of sliding in which sliding is assumed to cover the entire boundary rapidly but stops at the triple junction which requires accommodation, in the present approach the triple junction in a material of grain size 5.0 μm, for example, can only be reached after more than 3000 basic units have sheared sequentially. Thus, the stress buildup at the triple junction is long range and is analogous to the stress concentration that develops at a dislocation pile up (in dislocation theory) that helps transfer deformation from one grain to the next. The details of plane interface formation in mesoscopic sliding will depend on whether the interface is intercrystalline or interphase. An estimation of the threshold stress τ_0 for plane interface formation involving intercrystalline as well as interphase boundaries is given in Appendix 2. No specific shape of the grains is assumed because mesoscopic boundary sliding (and hence structural superplasticity) are independent of grain shape.

It is evident that the long range threshold stress in the case of single phase superplastic alloys of uniform grain size varies as $L^{-0.5}$ (equation (37)). In two phase and multiphase materials, both intercrystalline and interphase boundaries are present and their proportion will also vary from one material to the next. Thus, the threshold stress, which for this case is governed by both equations (37) and (38), will vary as L^{-a} , where a is a system and test condition dependent positive constant.

In view of equation (36), it can be written that

$$\tau_0 = \left(\frac{2\mu\gamma_B}{\alpha_f} \right)^{1/2} \left(\frac{\Delta A/A}{(NA)^{1/2}} \right)^a \quad \dots \quad (21)$$

where γ_B is the specific grain boundary energy, α_f is a form factor of the order of unity, and the constant a will have to be determined experimentally. All that can be said at this stage is that a will lie between 0 and 1 (the upper limit is fixed by equation (38)).

Evidently, when testing is in the tensile mode τ and τ_0 should be replaced by σ and σ_0 (the tensile equivalents of τ and τ_0) in the above equations and γ_0 and $\dot{\gamma}$ by ϵ'_0 and $\dot{\epsilon}$, respectively.

Optimal superplastic deformation

It is evident from the analysis presented in the section on cooperative/mesoscopic sliding above, that to describe optimal superplastic flow, the quantity τ in equations (17) to (20) (derived for predicting the sliding rate at a single boundary) should be replaced by $(\tau - \tau_0)$ (or in the tensile mode σ should be replaced by $(\sigma - \sigma_0)$). Let $\sigma' = (\sigma - \sigma_0)$. Assuming a two parameter lognormal distribution function for $g_1(\sigma'_i)$, $(g_1(\sigma'_i)$ and $f_1(\sigma'_i)$ are the functional forms of $g(\sigma_i)$ and $f(\sigma_i)$ when they are expressed in terms of the new variable σ')

$$g_1(\sigma'_i) = (\sigma'_i \lambda \pi^{1/2})^{-1} \exp \left\{ - \left[\frac{1}{\lambda} \ln \left(\frac{\sigma'_i}{\sigma_m} \right) \right]^2 \right\} \quad \dots \quad (22)$$

where σ_m is the mean and λ is the standard deviation. A lognormal distribution is assumed because of its common use in problems of this kind.

From equations (19) and (20)

$$\frac{df_1(\sigma')}{d\sigma'} = \int_0^{\sigma'} g_1(\sigma'_i) d\sigma'_i \quad \dots \quad (23)$$

Therefore

$$\frac{df_1(\sigma')}{d\sigma'} = \int_0^{\sigma'} g_1(\sigma'_i) d\sigma'_i = \Psi \left(\frac{1}{\lambda} \ln \frac{\sigma'}{\sigma_m} \right) \quad \dots \quad (24)$$

where

$$\Psi(x) = \frac{1}{\pi^{1/2}} \int_{-\infty}^x \exp(-x^2) dx$$

Since Ψ can not be obtained by integration, the following approximations are used.

For $0 \leq \sigma' \leq \sigma'_1$

$$\Psi = 0 \quad \dots \quad (25a)$$

For $\sigma'_1 \leq \sigma' \leq \sigma'_2$

$$\Psi = \frac{1}{2} + \frac{1}{\lambda \pi^{1/2}} \ln \frac{\sigma'}{\sigma_m} \quad \dots \quad (25b)$$

For $\sigma'_2 \leq \sigma'$

$$\Psi = 1 \quad \dots \quad (25c)$$

where

$$\sigma'_{1,2} = \sigma_m \exp\left(\mp \lambda \frac{\pi^{1/2}}{2}\right) \quad \dots \quad (26)$$

integration of equation (25b) in the region of interest $\sigma'_1 \leq \sigma' \leq \sigma'_2$ yields

$$f_1(\sigma') = \sigma' \left[\frac{1}{2} + \frac{1}{\lambda \pi^{1/2}} \left(\ln \frac{\sigma'}{\sigma_m} - 1 \right) \right] + \frac{\sigma'_1}{\lambda \pi^{1/2}} \quad \dots \quad (27)$$

Combining equations (24) and (27) gives

$$\frac{d \ln f_1(\sigma')}{d \ln \sigma'} = \frac{\sigma' \left[\frac{1}{2} + \frac{1}{\lambda \pi^{1/2}} \ln \left(\frac{\sigma'}{\sigma_m} \right) \right]}{\sigma' \left[\frac{1}{2} + \frac{1}{\lambda \pi^{1/2}} \left(\ln \frac{\sigma'}{\sigma_m} - 1 \right) \right] + \frac{\sigma'_1}{\pi^{1/2}}} \quad \dots \quad (28)$$

Let $\alpha = 2/\lambda \pi^{1/2}$ and σ' be expanded linearly as $\sigma' = \sigma_m + u$. Then (see equation (20))

$$n = \left(\frac{d \ln \dot{\epsilon}_{sp}}{d \ln \sigma} \right)_{T,L} = \left(\frac{d \ln f_1(\sigma')}{d \ln \sigma} \right)_{T,L} = \left(\frac{1}{1 - \alpha \left(1 - \frac{\sigma'_1}{\sigma_m} \right)} \times \left\{ 1 + \frac{u}{\sigma_m} \left[1 + \alpha - \frac{1}{1 - \alpha \left(1 - \frac{\sigma'_1}{\sigma_m} \right)} \right] \right\} \times \frac{\sigma}{(\sigma - \sigma_0)} \right) \quad \dots \quad (29)$$

which is an important practical parameter in superplasticity, and $\dot{\epsilon}_{sp}$ is the strain rate of superplastic deformation. From equations (20) and (27) the strain rate during optimal superplastic flow is given by

$$\dot{\epsilon}_{sp} = 2\omega_0 \frac{W}{L} \frac{\epsilon_0^2 V_0}{kT} \times \left\{ (\sigma - \sigma_0) \left[1 + \alpha \left(\ln \frac{\sigma'}{\sigma_m} - 1 \right) \right] + \sigma'_1 \alpha \right\} \quad \dots \quad (30)$$

Equation (30) is transcendental in α and can be solved by a trial and error procedure.

As $\sigma' = \sigma_m + u$ and

$$\ln \frac{\sigma'}{\sigma_m} = \ln \left(1 + \frac{u}{\sigma_m} \right) \approx \frac{u}{\sigma_m}$$

it follows that

$$\sum_{i=1}^{N_1} \left[\dot{\epsilon}_{spi} - \frac{C_1}{\sigma_m} \left\{ (\sigma_i - \sigma_0) [\sigma_m + \alpha(\sigma_i - \sigma_0 - 2\sigma_m)] + \sigma_m^2 \exp \left(-\frac{1}{\alpha} \right) \alpha \right\}^2 \right] = \Phi(C_1, \sigma_0, \alpha, \sigma_m) \quad \dots \quad (31)$$

where

$$C_1 = \left(2\omega_0 \frac{W}{L} \frac{\epsilon_0^2 V_0}{kT} \right)$$

and Φ is a function of C_1 , σ_0 , α , and σ_m , and N_1 is the number of experimental measurements of $\dot{\epsilon}_{sp}$ versus σ combinations. Evidently, the best fit values of C_1 , σ_0 , α , and σ_m are obtained by minimising the function Φ .

Once σ_0 , σ_m , and λ are known, $f_1(\sigma')$ for each temperature can be determined using equation (27), and $g_1(\sigma')$ can be

determined from equation (22) by replacing σ'_i by σ' . The functions $g_1(\sigma')$ and $f_1(\sigma')$ have been computed for a few superplastic alloy systems and are reported in Part 4. The method for determining the true activation energy for the rate controlling process is given in Appendix 3.

Grain growth/microstructural evolution during superplastic flow has not been taken into account in the present analysis. Thus, for the present, L may be viewed as an appropriate grain size value that has to be used in the equations developed above for an accurate fit of the experimental data. However, it is seen in Part 4 that as grain growth during superplastic deformation is relatively small (compared with the overall increase in the specimen length), excellent predictions result even when L is taken to be the initial grain size.

Discussion

The present paper has attempted to explain the origin of regions I and IIa of superplastic flow, i.e. up to the point of inflection in the sigmoidal $\ln \sigma - \ln \dot{\epsilon}_{sp}$ curve, in terms of a single boundary sliding controlled deformation process. To date no rigorous analysis is available for predicting the stress at which region IIb of superplastic flow will commence, as a function of grain size and temperature. This may form the subject of a future investigation.

As the initial boundary structure may not always be conducive to easy sliding, during early deformation the positions of the atoms in the basic units rearrange so that further sliding becomes easier. This would suggest an initial unsteady region of flow, as predicted by a previous, simplified analysis.⁶⁵

By comparing the microstructural characteristics of regions I and IIa, Vatsava and Langdon⁶⁶ and others (for a summary, see Ref. 18) have concluded that the contribution from sliding to the total strain is less in region I than in region II. As the strain rates are lower in region I than in region IIa, more time is available for boundary migration in region I. Consistent with this, greater boundary migration is observed experimentally in region I. This has led to the conclusion that the contribution of boundary sliding is less in region I than in region IIa, as it is often assumed that the contributions from sliding and diffusion (migration) are additive.

The activation energy in region I of superplastic flow is often reported to be greater than that in region IIa. However, it is demonstrated in Part 4 that when the stress and temperature dependences of the stress exponent n are not ignored, it is possible to assign a single true activation energy value to both regions I and IIa.

From equation (20), an expression for the grain boundary viscosity η can be obtained by noting that

$$\eta = \frac{\tau}{\dot{\gamma}}$$

That is

$$\eta = \tau \frac{1}{v} \left(\frac{L}{W} \right) \left(\frac{kT}{\gamma_0^2 V_0} \right) \frac{1}{f(\tau)} \exp \left(\frac{\Delta F_0}{kT} \right) \quad \dots \quad (32)$$

In Appendix 4 the variation in the viscosity of a superplastic alloy with applied stress, temperature, and grain size is considered.

Following the present authors' original proposals concerning grain/interphase boundary sliding controlled flow,⁴ there was an exchange⁶⁷⁻⁶⁹ with Gifkins and Langdon in which nine points of disagreement were identified. In the intervening years, Langdon and co-workers seem to have accepted the view that there is an initial unsteady region during superplastic flow and that nearly 100% of the external strain is caused by grain boundary sliding.⁷⁰⁻⁷³ In

earlier work^{4,68} the present authors had not identified the details of the micromechanism of sliding which made some assumptions *ad hoc* at best. An effort has been made in this paper to set this shortcoming right. The undesirability of converting the grain boundary width into an adjustable parameter, as in the 'core and mantle' analogy, has been realised by many and there is general acceptance of the view that cavitation/void formation need not be a concomitant of the optimal superplastic deformation process. Using equation (20) the sliding rate of any given random high angle grain boundary can be predicted.

An examination⁷⁴ of the alternative methods available for determining the activation energy for the rate controlling process appears to have endorsed these views, and along with Ashby and Verrall,³⁹ the present authors believe that regions I and IIa arise from the same physical process. This approach has the advantage of reducing the number of mechanisms needed to explain superplastic flow by one.

A distinction is made between the models of superplasticity that lead to an equation of flow similar to the Dorn equation and the Dorn equation itself, using which two empirical values are assigned to the stress exponent n in regions I and II to obtain two different activation energies for two unidentified processes. Conversely, the models of superplasticity assign specific values to n , and where applicable also predict its variation with the experimental conditions. Thus it would appear that the earlier controversy⁶⁷⁻⁶⁹ may be regarded as settled.

By suggesting that mesoscopic sliding is facilitated by local migration brought about by diffusion at intercrystalline boundaries and dislocation motion at interphase interfaces, the present model is able to account for dominant sliding, and limited diffusion and dislocation motion that have been observed experimentally. The speculation⁶² that lattice dislocations originally traverse the grains and become absorbed at the boundaries conferring on the boundaries fluidlike characteristics beyond a minimum stress level is also unnecessary, as the present approach can account for the low values of m ($=1/n$) in region I (equation (29)) and the observation of non-rate controlling dislocation activity in the optimal range of superplastic flow, without making such an unsubstantiated assumption. In fact, the present analysis would predict that in the optimal range of superplastic flow, evidence for dislocation activity will be present in those grains contributing to boundary sliding significantly which are surrounded by at least one active interphase boundary. Experimental evidence of this is presented in Part 2.

Conclusions

1. A careful examination of the experimental results concerning optimal structural superplasticity suggests that the phenomenon is the result of grain/interphase boundary sliding-diffusion coupled flow.
2. A model that postulates that boundary sliding aided by local migration (to facilitate mesoscopic boundary sliding) is the rate controlling step in superplastic flow in both regions I and IIa is presented. Equations have been derived which predict the strain rate of deformation as a function of applied stress, temperature, and grain size.
3. The correct procedure for evaluating the true activation energy for the rate controlling process has been given.
4. The reported differences in the microstructural and phenomenological features of deformation in regions I and IIa of superplastic flow are shown to be apparent.
5. Expressions have been derived for the internal stress distribution arising from boundary sliding, the long range threshold stress associated with mesoscopic sliding, and the viscosity of a grain boundary as well as that for a

superplastic alloy. The presence of an initial unsteady region of flow, predicted in an earlier paper,⁶⁵ is confirmed.

Appendixes

Appendix 1

Consider Fig. 4, which is derived from Fig. 2. Only the two atoms immediately above and below the central atom (e.g. atom marked 1, 2 or 3 in Fig. 2) are considered to cover the grain boundary width, the full and broken lines representing the equilibrium configuration C_{eq} and the topologically equivalent configuration C'_{eq} , respectively, and S_0 is the local displacement of the atoms in the basic unit corresponding to a local shear γ_0 .

If all basic units in the boundary experience a shear γ_0 , then the two grains forming the boundary would be displaced by S_0 relative to each other. The macroscopic shear γ , therefore, is S_0/L , where L is the grain size. Thus, $\dot{\gamma} = \dot{S}/L$, where the 'dot' indicates a time derivative. Further, from Fig. 4 one may write by inspection

$$\sin \gamma_0 = \frac{S_0/2}{b} \approx \gamma_0$$

where b is the interatomic distance. But $W = \sim 2b$, which leads to

$$S_0 \approx W\gamma_0$$

Appendix 2

The threshold stress needed to be overcome before a plane interface that allows mesoscopic sliding can form, is estimated below. The results depend on whether intercrystalline or interphase boundaries are involved.

MESOSCOPIC SLIDING AT INTERCRYSTALLINE BOUNDARIES

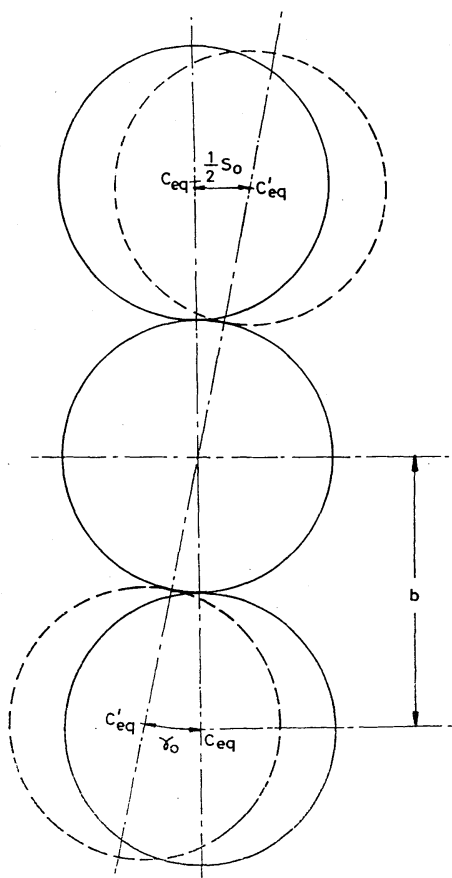
Schlipf⁶¹ has considered several situations which will be reported elsewhere. In the present work the polycrystalline structure is described as a cubic dense packing of equisized spheres. Let the walls of the Wigner-Seitz cell containing one sphere represent the grain boundaries. This corresponds to a face centred cubic structure on a mesoscopic scale with glide planes and glide directions. Since the roughness is the lowest on the $\{111\}$ planes, these are assumed to be the glide planes for long range sliding. The surface of a $\{111\}$ plane in the mesoscopic structure consists of a regular arrangement of peaks and troughs whose height h will vary as the average grain size L regardless of grain shape. Likewise, the ground area A will vary regardless of grain shape as L^2 .

In order to create a plane interface, the peaks and troughs of the mesoscopic $\{111\}$ plane must be removed, i.e. the grain boundary area in the mesoscopic glide plane must be decreased by local migration. This is possible only if the grain boundary area perpendicular to this plane increases. Let the net increase per grain be ΔA , the number of grains participating in a mesoscopic sliding event N , and the specific grain boundary energy γ_B . Then the work done in creating the extra grain boundary area perpendicular to the mesoscopic glide plane ΔW_1 is given by

$$\Delta W_1 = N\gamma_B \Delta A \quad \dots \quad (33)$$

Conversely, on sliding through a distance ΔS under a shear stress τ_0 the external force does work ΔW_2 of

$$\Delta W_2 = \tau_0 A N \Delta S \quad \dots \quad (34)$$



4 Schematic representation for calculating shear and shear rate associated with unit process leading to grain/interphase boundary sliding

However, half this work is stored by transient elastic distortions along the border of the sliding area. Therefore, only $\Delta W_3 = \Delta W_2/2$ is available for supplying the energy ΔW_1 . In general, the sliding area will not extend to the surface of a test specimen. Thus ΔS must be accommodated elastically along the border of the slid area. These distortions will eventually be relieved by similar sliding processes in the neighbourhood and also by diffusional accommodation.

The linear dimension of the slid area may be written as $\alpha_f(NA)^{1/2}$ where α_f is a form factor of the order of unity. Thus

$$\Delta S = \alpha_f \tau_0 (NA)^{1/2} / \mu \quad (35)$$

where μ is the shear modulus. Combining equations (33)–(35) and equating ΔW_1 and ΔW_3 , gives

$$\tau_0 = \left(\frac{2\mu\gamma_B \Delta A/A}{\alpha_f (NA)^{1/2}} \right)^{1/2} \quad (36)$$

Evidently, τ_0 gives rise to a long range threshold (shear) stress.

At least to a first order approximation $(\Delta A/A(N)^{1/2})$ is independent of grain size. So we represent $(\Delta A/\alpha_f(N)^{1/2})$ as a constant β_f whose magnitude will depend on the grain shape assumed. If $A = \gamma_f^2 L^2$, where γ_f is a grain shape dependent constant, it follows that

$$\tau_0 = \left(\frac{C_f}{L} \right)^{1/2} \quad (37)$$

where $C_f = (2\mu\gamma_B\beta_f/\gamma_f)$. Thus, in this approach, in material of uniform grain size L the long range threshold stress arising from mesoscopic sliding at intercrystalline boundaries varies as $L^{-1/2}$.

MESOSCOPIC SLIDING AT INTERPHASE BOUNDARIES

Interphase boundaries of phases differing significantly in composition can not migrate without material transport by diffusion over distances of the order of the grain size. Here, the plane interface has to be created by non-rate controlling plastic deformation of grain corners in a manner similar to that suggested by Perevezentsev *et al.*,⁶² with the difference that in the present approach the residual misfit in the boundary after the operation of this process (which will be a fraction of the lattice Burgers vector) is removed by local migration, as discussed above.

Thus, at interphase interfaces the residual misfit is independent of grain size and ΔA varies with L , but A varies, as before, with L^2 . This leads to the prediction (from equation (36))

$$\tau_0 = \left(\frac{C_{f1}}{L} \right) \quad (38)$$

where C_{f1} has absorbed all other constants.

Appendix 3

The true activation energy for the rate controlling process is determined as follows. From equations (12) and (20) (after replacing σ by σ') it follows that for a material of constant grain size

$$\ln \dot{\epsilon}_{sp} = \ln K_1 + \ln v - \ln(kT) + \ln[f_1(\sigma')] - \frac{\Delta F_0}{kT} \quad (39)$$

where $\dot{\epsilon}_{sp}$ is the strain rate of superplastic deformation and where

$$K_1 = \left(\frac{2W}{L} \epsilon_0^2 V_0 \right)$$

$(\ln K_1 + \ln v - \ln(kT) - (\Delta F_0/kT))$ is equal to constant C_1 (see section on optimal superplastic deformation above) for a given temperature and using a set of $\dot{\epsilon}_{sp}$ versus σ values corresponding to each temperature this constant C_1 can be evaluated for each temperature by a least squares analysis (equation (31)). Then, using these values of C_1 and the values of v (equal to either kT/h (Ref. 63) or a constant of magnitude 10^{13} s^{-1} (Ref. 64) and Boltzmann's constant, the following equation can be written

$$(C_1 - \ln v + \ln kT) = \ln K_1 - (\Delta F_0/kT) \quad (40)$$

But $\Delta F_0 = \Delta U_0 - T\Delta S$ where ΔU_0 represents the true activation energy and ΔS is the activation entropy which will be included in the fitting parameters. Therefore

$$(C_1 - \ln v + \ln kT) = \ln K_2 - (\Delta U_0/kT) \quad (41)$$

where

$$\ln K_2 = \ln K_1 + \frac{\Delta S}{k}$$

Thus K_1 , K_2 , and ΔU_0 are expected to be independent of temperature but C_1 and ΔF_0 will be temperature dependent.

The true activation energy ΔU_0 for the rate controlling deformation process is evaluated from equation (41).

In Part 4 the true activation energies for a few superplastic alloy systems will be reported.

Appendix 4

Here the dependence of the viscosity of a grain boundary in a superplastic alloy on stress, temperature, and grain size is considered.

VARIATION OF BOUNDARY VISCOSITY WITH TEMPERATURE, STRESS, AND GRAIN SIZE

Temperature dependence

When the grain size and stress are constant equation (32) becomes

η = C_3 \frac{kT}{v} \exp\left(\frac{\Delta F_0}{kT}\right) \dots \dots \dots (42)

where C_3 has absorbed all other constants. Therefore, for the two situations of v = kT/h and v = 10^{13} s^{-1} = constant

\frac{\partial \eta}{\partial T} = C_4 \left(\frac{\Delta F_0}{k}\right) \left(-\frac{1}{T^2}\right) \exp\left(\frac{\Delta F_0}{kT}\right) \dots \dots \dots (43a)

and

\frac{\partial \eta}{\partial T} = C_5 \left[1 - \frac{\Delta F_0}{kT}\right] \exp\left(\frac{\Delta F_0}{kT}\right) \dots \dots \dots (43b)

where C_4 = C_3 h and C_5 = C_3 k/h, respectively. As C_4, C_5, and exp(\Delta F_0/kT) are all positive and (\Delta F_0/kT) > 1 (see Part 4 for experimental results), it follows from both equations (43a) and (43b) that the viscosity decreases with increasing temperature.

Stress dependence

When the grain size and temperature are constant, from equations (32) and (27) (with σ or τ replacing σ' or τ')

η = C_6 τ \frac{1}{\left\{ \tau \left[\frac{1}{2} + \frac{1}{\lambda_s \pi^{1/2}} \left(\ln \frac{\tau}{\tau_m} - 1 \right) \right] + \frac{\tau_1}{\lambda_s \pi^{1/2}} \right\}} \dots \dots (44)

where C_6 has absorbed all other constants. Thus

\frac{\partial \eta}{\partial \tau} = \frac{C_6}{\tau \left[\frac{1}{2} + \frac{1}{\lambda_s \pi^{1/2}} \left(\ln \frac{\tau}{\tau_m} - 1 \right) \right] + \frac{\tau_1}{\lambda_s \pi^{1/2}}} \times \left\{ 1 - \frac{\frac{1}{2} + \frac{1}{\lambda_s \pi^{1/2}} \left(\ln \frac{\tau}{\tau_m} + \tau_m - 1 \right)}{\tau \left[\frac{1}{2} + \frac{1}{\lambda_s \pi^{1/2}} \left(\ln \frac{\tau}{\tau_m} - 1 \right) \right] + \frac{\tau_1}{\lambda_s \pi^{1/2}}} \right\} \dots \dots (45)

where τ_m is the mean and λ_s is the standard deviation of the internal shear stress distribution at a single boundary and τ_1 = τ_m exp(-λ π^{1/2}/2). Whether the viscosity will decrease or increase with increasing stress will depend on whether the expression on the right hand side of equation (4.4) is negative or positive.

Grain size dependence

When the temperature and stress are constant, it follows from equation (32) that η ∝ L. That is, the boundary viscosity η increases linearly with grain size.

Superplastic alloy

Equation (30) is the basis for deriving the dependence of viscosity on the variables of temperature, stress, and grain size. A procedure similar to that adopted above will be followed: σ_0, σ_m, and α change with both temperature and grain size. In the tensile mode η = (σ'/3ε̇_sp). As the calculations in this case follow the same pattern, it is not considered necessary to set them out here.

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